



Examining the variables that impact Student satisfaction at AASTMT.

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Abstract:

This research studies various factors that affect student satisfaction at the Arab Academy for science, technology and maritime transport (AASTMT). As students satisfaction is a critical key performance indication (KPI) for measuring the service quality in the higher educational institutions. The study identifies four primary variables that influence students satisfaction that being, students expectations, academic services, campus infrastructure and previous recommendations

Methodology:

The study implemented a correlational research design with a cross-sectional time horizon meaning that the data was collected only once through out the whole research. The data was collected through online questionnaires from a diverse sample of 76 students distributed across various AASTMT branches all across Egypt, including both as respondents local and international students. The study used a 5- point liker scale in which the participants chooses to what extent he can agree or disagree with the studied variable, it was later on coded and validated using Cronbach's alpha, with campus infrastructure score the highest reliability of 0.896.

Introduction section:

Problem statement:

Student satisfaction is one of the most important KPIs used by university and higher education institutions to measure the quality of their services. Despite the increasing expenditure on higher education, there's a move towards dropping out or even skipping this educational phase as some may comment that it's a higher price for a lower value.

Acknowledging that price does play a crucial role in determining the overall student satisfaction level, there are other factors worth mentioning such as student expectations, academic services, campus infrastructure, employability skills and previous recommendations. Data will be collected from students at Arab Academy for science and technology and martime transport to explore these factors to examine the overall student satisfaction at AASTMT.

Research objectives:

1. To analyze the affect of student expectations on student satisfaction.
2. To examine the influence of academic services on the overall student satisfaction.
3. To study the role of campus infrastructure on student satisfaction
4. To investigate the impact of previous recommendations on student satisfaction.

Research questions:

1. What is the affect of campus infrastructure on student satisfaction?
2. How does student expectations influence the overall student satisfaction?
3. How does academic service impact student satisfaction?
4. How do previous recommendations affect student satisfaction?

Literature Review:

Student Expectation:

Student expectations represent a user's belief about the likelihood that a product or service will bring them certain benefits and outcomes. Student expectations act as a benchmark in which students can compare their actual academic experience to their perceived expectations. When universities meet or exceed their expectations, satisfaction increases, the same when they fail to meet their expectations then satisfaction will decrease. Other studies confirm that student satisfaction is a complex construct of many contributing factors, such as previous expectations mentioned by family members and friends or even what they've perceived from the media portrayal of the education institution [(Villafaina & Mosquera, 2024)]. All these factors help in forming the students' expectations. [Tomlinson et al., 2023].

However, some studies may not totally agree with the above statements, as others may describe the relationship between students' expectations and students' satisfaction in another form. The study shows that determining the student expectations depends on the course form, whether it's a traditional course or a modern course form, with the modern of course being newly implemented ways of learning such as online courses, vice versa traditional courses being the ones held face-to-face or in campus. The study goes on to contradicting its own cause which was based on the hypothesis that less traditional courses or less face-to-face decrease the student expectations which lead to higher student satisfaction, which was later proved wrong as a new variable was presented that influences the student expectations called 'challenge'. The first hypothesis didn't work because when the less face-to-face courses a student experience, he expects less and less workload which in return put out that fire of challenge inside them, eventually the students will start to believe that the presented course isn't worthy of their time, one thing led to another and suddenly students' satisfaction drops due to the shortage in firewood that students' expectations feeds upon. Observing from the above statements that the presence of challenge mediates the positive relationship between student expectations and student satisfaction.

The story doesn't end here, since the development of the first hypothesis failed, the study identified a new hypothesis that may be the alternative for the first one. In the first hypothesis we can conclude that a wide gap has been identified between students' expectations and students' satisfaction, we've also the variable influencing this gap which was the academic challenge perceived by the students which later on determined how they perceive the presented course or workload. The newly developed hypothesis doesn't necessarily solve or fill in the gap between students' expectations and students' satisfaction but we can describe it more as a hypothesis that can adapt to that gap or at least shorten the in-between gap. The hypothesis basically removes the traditional course form from the equation and suggests that the modern courses form is enough to put on that workload stress and challenge on the students which in return increases the overall students' satisfaction.

In other words, what happened here is that the workload stress or perceived challenge by the students towards the presented course doesn't necessarily diminish the gap but in another way it just shortens the gap between students' expectations and students' satisfaction. It doesn't end here, the study goes on in identifying another factor that goes on by the name reality or as mentioned in the study as the perceived quality. Reality here is just how the student highlights or perception of the courses, which is of course formed by the student's previous experience in the overall courses form either it a traditional course form, that face-to-face courses or if its modern courses, that of online courses. If we took a closer look at the relationship between students' expectations and student satisfaction, we'll find that reality here plays a significant role, as it acts as a moderator between the students' expectations and student satisfaction.

With that the whole equation is complete, with the presence of that challenge factor mediating the students' expectations along side with crucial role played by the reality factor in moderating the whole relationship between the students' expectations and the students' satisfaction. Congrats, the gap has been solved, but how exactly did we reach to that solution? Let's break it down, addressing first the moderator, reality, or as mentioned in the conducted study as the perceived quality, its role was to basically describe the relationship, either it was high enough to meet the students' expectations or that the expectations were aligned perfectly to meet the perceived quality. The perceived quality can be shown in many different forms in the university, such as exceptional professors, up to date technologies or even highly engaging courses. The study shows that the reality factor plays a more important role rather than the mediating factor, the challenge, it goes on to stating that a high perceived quality can influence directly the students' satisfaction regardless of the initial gap. That's why we can state that the main cause for the initial gap was actually the absence of that mediating factor which was the perceived challenge by the students.

That's why higher educational institutions must set clear expectations in advance that serve an antibiotic that prevents that challenge of forming in the first place. If the students' expectations is set to align perfectly with reality, then by default the dual relationship will move into harmony, one thing led to another and suddenly the gap is

solved. Breaking down the mediator we can clearly suggest that the most logical method of solving the gap was just to eliminate the mediator from the whole equation. But how can the challenge be eliminated? By the presence of experience, when a student has enough experience he can easily adjust his judgments or certain expectations towards the course, its crucial cause is some cases you can encounter students that may not have experienced any form of that online courses, leading to them failing in aligning their expectations to the reality of the course. Leading them to eventually tilting towards the challenge factor once again in order to help them set their expectations, we that we can already see that the gap has been formed again.

That's why that factor of experience is crucial as it acts as a deactivator for the mediator preventing any formation of the gap. However with all the stated above, a question arises, how can students gain experience? Here comes the role of the higher educational institutions of providing actual workshops or training to new students for example a time management Training. This training directly addresses the root cause of the problem caused by the challenge, by replacing the misinformation with accurate information, thus breaking that mediating chain.

In conclusion, the gap is solved when the mediator is broken(challenge) through students experience. Allowing the students expectations to align perfectly with the reality or perceived quality, which in return achieves that desired outcome of harmony relationship and stable students satisfaction.[(Pinto & Anderson, 2013)]

Academic Services:

Conversely, there's a broad agreement that academic services or perceived quality of service have been playing a significant role in determining the students satisfaction for many years. Service quality has been defined as the overall impression a consumer or customer has about a certain product or service. The study goes on further into explaining that service quality is divided into two aspects, tangible and intangible aspects or in other words, academic and non-academic aspects. The academic aspect focus more on evaluating the educational qualities based on student feedback and curricular analysis of the courses itself, it may include factors such as the expertise of the professors or even the ability from an administrative side to come up with a solution for problems that students face. On the other hand, there are the non-academic aspects focus on the physical educational atmosphere in the university, such as the quality of facilities, classrooms and equipment provided by the campus.

However, going further on with the study, a new variable defines itself as the perceived employability skills. Which is the development of knowledge and skills needed to be able to perform a job well following the studies undertaken. Or in other words its more of a future utility, when students perceive their education as a long term investments. At first it looks like it acts as a mediator for the relationship between the service quality and students satisfaction, but it's the complete opposite. It presents itself as a new variable that impacts the student satisfaction directly, along side with perceived service quality becoming a mediator for the relationship between the employability skills and the students satisfaction. The perceived employability skills is just one of many factors such as the professors skills or campus facilities that feeds into that mediating factor [(Villafaina & Mosquera, 2024)].

Other studies may mention a slightly different approach to the problem, the study starts by stating various factors that affect the students satisfaction such as the increased competition between higher educational institutions, demographic changes and the increased educational costs. Which led them to develop an approach backed the theory of stakeholders, naming the approach as the student as a customer or SAC. Meaning that with the increased competition between higher education institutions, universities should treat their students as stakeholders that belong to their institutions and therefore universities should thrive into achieving students satisfactions in order to hold and maintain those so called stakeholders as long as possible.

Understanding from the above statements that in the current educational economical situation, students satisfaction is the main goal and target of higher education institutions. Metaphorically we can say that in this scenario, the universities will be a train that runs on coal which is students satisfaction. The model has both a positive and negative consequences, a positive consequence in which universities will do their best

to meet students satisfaction such as scheduling classes into time periods that may be more than convenient, suitable and even desirable for students. In addition to that universities may start to listen more into students demands and implement them such as increasing off-class activities. They can also go to the extent of offering a variety of courses available.

All these factors can lead to and increase in students satisfaction and make it a win-win situation for both students and the university itself. However that's not always the case, remembering that the study mentioned students as stakeholders and the whole educational sector as a market, it is widely known that in any market sector there is the profit and loss. In this case however the loss does not affect any of the both sides but instead it has a direct affect on the sole purpose of the whole research, which is the educational standards, even though the approach may increase students satisfaction in the universities, on the other hand this increase comes with a huge cost on the educational standards, in the long-term this may affect the whole educational systems.

But why is that going to happen? In the beginning of the study, they mentioned that there's an increased competition in the educational market in addition to a wide change in demographics, lets discuss every point in details, the sole purpose of the educational sector is to turn students into qualified individuals who are able to work and perform in different sectors of the economy, think of it as a huge garden, when new plants(that's other sectors) start growing fast and spreading, they attract more bees, sunlight and attention, simultaneously they also push nearby plants to grow faster to survive(that's competition). Education being one of these plants suddenly finds itself competing for same resources in order to survive and grow taller. The more the growth around it, the more the competition within it. Understanding that the main reason there's an increasing competition in educational sector is because there's an increase in competition in other sectors of the economy as well. Linking the competition to the context, there's an increase in competition cause resources are going scare and markets needs employees that can do double the work needed with the same effectiveness and efficiency, which puts a huge pressure on the educational sector on matching the market preferences and work criteria then translating it educational methods that can be implemented by universities.

Returning to the main point that when universities thrive more and focus more on achieving students satisfaction it can have a negative impact on the long run on educational sector which in return will have a negative impact on the other sectors of the economy. Apart from that, it can have other negative consequences, such as forming a grade inflation, as where universities will go easy on grades such as promoting easier grading to satisfy students or go even to the extent of offering easier exams. Its true that in this case the university may achieve students satisfaction however its all going to be in the short-term, as in the long-term it's a completely different case, as with this easy grading systems, the university reputation may fall down and potential new students may not even consider joining or enrolling in any course offered by the university.

Furthermore, implementing the SAC model may have a negative impact on the students themselves without them realizing, changing the university system just in order to achieve or increase students satisfaction may come with a disadvantage on the students personality itself. A university main role is to qualify students to the open world, and that does not come with the academic factors it also comes with other socials factors that students must learn, understand and implement it on their own day-to-day lives. If a university offers classes scheduling to students in convenient time periods that students desire that will for a fact increase students satisfaction but in the same time reduce the students responsibilities or shift the responsibility of learning away from the students.[(Bordean & Sonea, 2018)]

Campus infrastructure:

So far in our research paper we have mentioned two intangible factors that have a impact on students satisfaction, that's students expectations and the academic services. Going on to our perhaps the only

tangible factor included in the research, campus infrastructure. The campus infrastructure plays a crucial role on determining the overall student satisfaction level, as it turns out that the only tangible factor in the whole equation is where we get to see and observe the other intangible factors such as students expectations, academic services and previous recommendations. The campus infrastructure can basically be defined as the physical place or location the students experience any form of education in, such as classrooms, training facilities, and the overall external environment. However, its been stated that the campus infrastructure plays another role that increasing or maintaining students satisfactions as it can also contribute to students retention. [(Villafaina & Mosquera, 2024)].

However, other studies suggest that campus infrastructure is rather made up of complex factors. The study states that campus infrastructure can be viewed as a tangible variable and an intangible variable simultaneously at the same time. Being considered as an intangible factor as per seeing the campus infrastructure as an experience lived by the students that contribute to their overall student satisfaction. The study sees campus infrastructure as a complex interacting system. Factors influencing campus infrastructure can be divided into two sections, physical environment which consists of tangible factors such as classroom, facilities, grounds, spaces and all sorts of buildings. Then there are the intangible factors or as mentioned in the study as the social environment which consists of community programs, support networks, internships and various activities. When those two factors are blended together and harmonized, then the end result would be a magnificent campus infrastructure that will certainly if not increase will maintain the students satisfaction in the university.

So far we can state that both the physical environment factors and the social environment factors act as a mediator in this relationship between campus infrastructure and the students satisfaction, as the both factor contribute and influence the campus infrastructure directly. The whole relationship between the two factors can be described as a chain, if one part of the chain is weak then all the other parts of the chain will weaken as well, if the instruments in this case the facilities and services are out of tune or non-functional then the whole composition of the students experience will suffer. In order to prevent that from happening, the environment either physical or social must be carefully and consciously in order to satisfy the student diverse needs of both physical and mental health, suggesting the need for the campus infrastructure to be dynamic, flexible for change and highly adaptable for all individuals within the higher educational institutions.

Furthermore with the understanding of the whole complex system, we can perceive it more as the biological ecosystems which requires a diverse combination of factors such as soil, water and sunlight, in the same way the academic ecosystems requires a clean, well maintained spaces and buildings, up to date technologies and supported services, the presence of all those factors is crucial in fostering to intellectual and personal development the students require on their educational journey. In the same way, a deficiency in this ecosystem such as poor buildings may act as a limiting factor, holding back the full potential of the students either physically or mentally, much like how a lack of nutrients will limit the crops yields.

The study goes on to more depth of the campus infrastructure, first stating the factors physical and social factors affecting the campus infrastructure, to now defining what campus infrastructure really is. Infrastructure is defined as the specific or functional facilities that provide mandatory support which later on enables the students for the academic and personal development. The study goes on to identifying various factors that play a crucial role in the overall campus infrastructure, such as library resources along side with quality study places which are framed as tools that encourage and foster a disciplined approach of studying, which in return will help in developing intellectual academic goodness or in other words, the whole higher education ecosystem will be healthy just because its based on a soil full of nutrients. Simultaneously, with the rapid growth of technology and it being integrated into the educational ecosystem more frequently, higher educational institutions must merge more reliable IT infrastructure into the educational experience such as implementing a stable WI-FI connection or even making online students portal. Other than meeting students expectations in the university, the higher educational institute will also benefit from this IT infrastructure implementation, as the data flow in the campus will become more reliable, more efficient and more effective, as it cuts down the paper usage and make the data flow more

sustainable.

In other words, we can perceive this functional or operational infrastructure system in the higher educational institutions just like the hydraulic system, whereas the system insures a smooth flow of data and information through the campus. When this water flow is blocked such as having a bad WI-FI connection or a lagging student portal, the whole system fails leading to a friction in operations. This friction decreases efficiency and lowers the overall performance in the campus, just as a poor hydraulic flow causes a machine to operate slowly or even fail.

The study goes on to acknowledging another factor as the institutional alignment or in other words, the administrative structure in the campus, it states that this factor should be unified and collaborative to effectively deliver the students experience. The study suggests that in order to achieve that supportive environment mentioned above as the harmonized combination of the physical and the social environment, there must be a harmony between academic affairs and the students affairs. This synergy ensures that campus facilities and services are accessible and truly supportive for the students. The academic affairs are responsible for everything related to teaching, learning and the academic quality, such as designing and approving academic programs and curricular. Scheduling courses, exams and academic calendars all the way to supporting research and academic development. On the other hand, the students affair are responsible for the students life, support and well-being outside the classroom such as students registration and enrollment in courses, academic advising and student guidance all the way to organizing student events or even leadership programs. In short, their responsibilities consists of how students are supported, guided and developed as individuals. When academic and students affairs are working together towards a common goal, it creates opportunities for supportive spaces which in return maximize the students engagement in the courses and their success. This harmony, forces the facilities to become a decisive factors in the equation, making the outcome dependent on the administrative ability to plan and design facilities that not just maintain but rather improve the students satisfaction in the campus. Concluding, from that the institutional alignment or the administrative ability acts as a moderator in the equation, as it being good or bad has a direct effect on the overall students satisfaction.

Concluding all the stated above, we can say that the university environment or in other words the educational ecosystem in the university which is divided into two factors that being physical environment and the other being the social environment, all act us as mediator for the campus infrastructure in its relationship with the overall students satisfaction. We've also defined our moderator that being the institutional alignment or the administrative structure of the campus which is the result of the harmony between the academic and the students affairs. With all that being said, we can state that a university cannot simply just provide resources, it must strategically align its regulations to ensure that the resources are accessible, supportive and most of all being efficiently and effectively used. The administrative structure is the key to the whole equation as it's the factor that transforms the investments provided by the higher educational institutions into and active outcome driven by support. [(Villafaina & Mosquera, 2024)].

Previous recommendations:

In the higher education sector, there's lots of competition between different universities which in return leads to students feeling overwhelmed and facing a hard time choosing which university to attend. To make a decision, they often look for previous recommendations, which essentially means the dissemination and exchange of comments and opinions, positive or negative, regarding the experience with a product or service, these comments and opinions often comes from people that they trust such as family members or friends. These previous recommendations plays a crucial role in determining the students satisfaction as its basically considered a starting point that helps in building the students initial view of the university.

In simple terms. Previous recommendations are just word-of-mouth (WOM) information that the students

get from their inner circles, think of a cousin telling you how their major was worth every penny or your friend warning you about a certain professor. according to the study, this isn't just a form of a casual normal chat its even more than that, it's a form of external communication the students entire perspective during the pre-purchase phase of the product as mentioned in the study. But why is that necessary in this whole process of choosing the university? The answer is quite simple, with the continuous rise in educational costs in the higher educational sector, students are forced to take those recommendations as precautions to protect their investment or in other words their long-term commitments as with this economical state the risk is high and continuous to increase. That's why students look for clues or hints from people whom they trust to reduce the risk of making any mistake.

The study goes on to explain the relationship between previous recommendations and not only students satisfaction but in addition to that student expectations. Stating that previous recommendations act as an antecedent or in other terms the starting point for the relationship between students expectations and students satisfaction, as apparently previous recommendations are considered the raw source of information and data that students collect and interpret which then forms their whole expectations towards the university. With that being said, we can state that the whole relationship between previous recommendations and students expectations can be summed up in one factor called influence. But what exactly is influence? Influence refers to the theoretical power that one factor has upon another factor whether its shaping it or even causing a change in it. When the research states that previous recommendations influence students expectations, it means that for every positive comment or opinion a students hears from either family members or friends, there's a predictable increase in the positive expectations students create and hold before starting university, and same process whenever there's a negative opinion present.

So far the research found out that previous recommendations do play a crucial role in determining the influence, however the study also finds out that the actual weight of the influence depends on the actual source of influence which vary from the students before enrolling in the university to the students after enrolling to the university. The study goes on to explaining the types of influence either it's the initial influence, which occurs when before the student actually enroll in the university, it based on previous recommendations being a primary source of influence, it basically that state when the student creates that mental image of the university from by the comments and opinions they received from their inner circles. The other type of influence will the secondary influence, which occurs once the student begin their actual studies, this type of influence is based on the perceived service quality that the students get once they step foot on the campus, it can vary depending on their perspective of the campus service and infrastructure itself, such as the quality of teaching and facilities. Which then forms a powerful influence final satisfaction towards the university than what they've been told.

From all the above, we can state that influence act as a mediator in both equations, either in the relationship between previous recommendations and students expectations, mentioned as the initial influence which is based on the recommendations students receive and interpret before enrolling. Then it being the mediator in the relationship between previous recommendations and students satisfaction, as mentioned above as the secondary influence which occurs after the student enrolls on campus, backed however by the perceived service quality instead of comments and opinions of family members and friends.

However, its important to note that the factor of previous recommendations depends highly on practical factors such as how far the university campus is from the student, or how much it costs. Or even on cultural factors, as different countries and cultures may have another point-of-view or perspective towards the whole higher educational sector. The presence of these factors are important in the whole relationship either it was between previous recommendations and students expectations or between previous recommendations and students satisfaction. As it determines and showcase the overall level of how a student perceive not just his/her university but instead the whole educational system. This means that while recommendations can be the vital spark and starting point of the whole journey, they are a part of a larger and more complex puzzle of decision making that can vary from student to student.

There's this also potential disadvantage along the way mentioned as, expectation contradiction where as if the recommendations the students hear where over exaggerated with this will lead them to setting high expectations and standards towards the university that may not be necessary present, which eventually leads to students forming this gap of confusion. But why will that gap occur? When the reality does not match the mental hype set by the student from comments he/her received, their satisfaction can actually drop faster from someone who had no expectations at all in the first place. That's why the study states that for the recommendations factor to be more effective and efficient, they must be reasoned by the university actual performance.

With the stated above, what exactly will be the take for universities? For universities, it means that their most effective salespersons aren't their staff at all but instead it's their students. By understanding and managing these previous recommendations, universities can build a stronger reputation for themselves and foster long-term sustainability, the equation to all that is quite simple, just keep the current students happy and in return they'll provide positive recommendations to the next satisfied generation of students.

Student Expectation

Expectations represent a user's beliefs about the likelihood that a product or service will bring them certain benefits and outcomes. [(Villafaina & Mosquera, 2024)].

Academic Services

The overall impression a consumer or customer has about a certain product or service. [(Villafaina & Mosquera, 2024)].

Campus Infrastructure

learning environment that promotes students physical and mental health and well-being. (Song et al., 2024)

Previous Recommendation

Previous recommendations are understood as the dissemination and exchange of comments and opinions, positive or negative, regarding the experience with a product or service. [(Villafaina & Mosquera, 2024)].

Hypotheses:

Students expectations:

H0=There is no relationship between student expectations and student satisfaction.

H1= There is a relationship between student expectations and student satisfaction.

H2=Higher student expectations are associated with lower student satisfaction.

H3=If student expectation increase, then student satisfaction will also increase.

Academic services:

H0=There is no relationship between academic services and student satisfaction.

H1=There is a relationship between academic services and student satisfaction.

H2=Better academic services are associated with higher student satisfaction.

H3=If the quality of academic services improves, then student satisfaction will increase.

Campus infrastructure:

H0=There is no relationship between campus infrastructure and student satisfaction.

H1=There is a relationship between campus infrastructure and student satisfaction.

H2=Better campus infrastructure is associated with higher student satisfaction.

H3=If campus infrastructure improves, then student satisfaction will increase.

Previous recommendations:

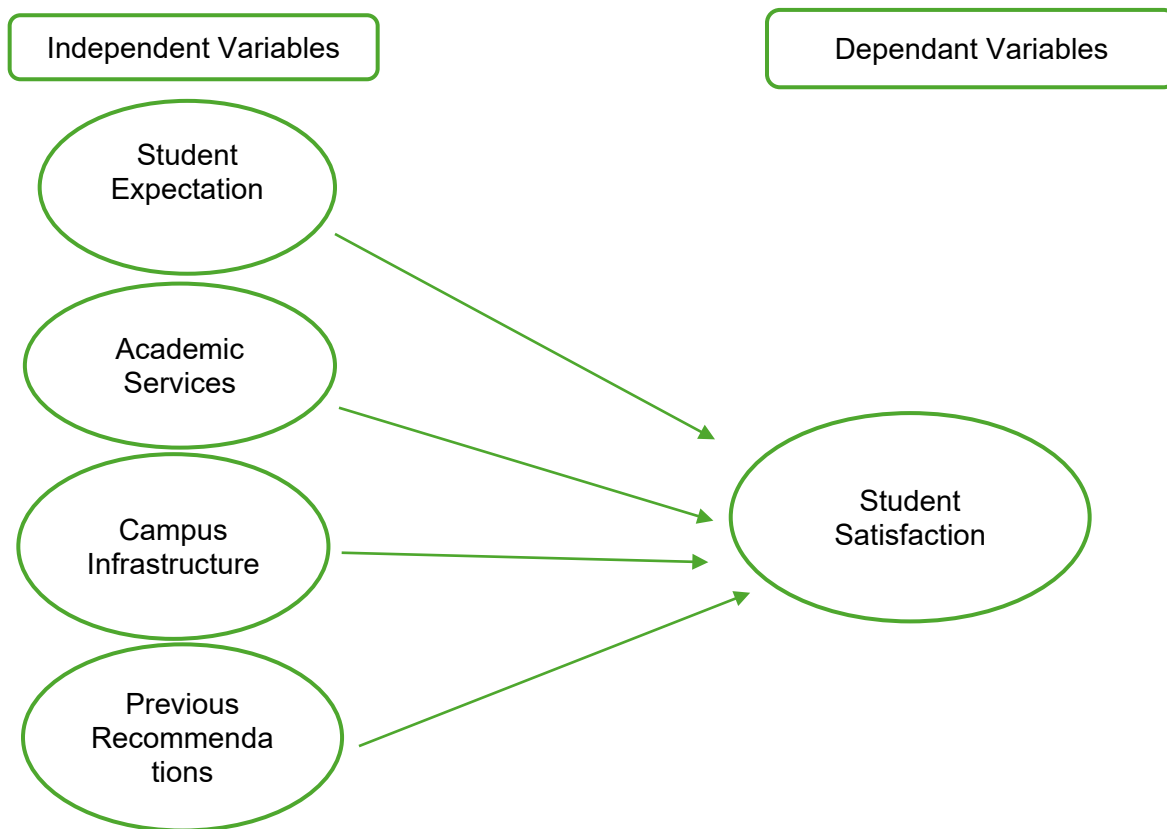
H0=There is no relationship between previous recommendation and student satisfaction.

H1=There is a relationship between previous recommendation and student satisfaction.

H2=Higher levels of previous recommendation are associated with higher student satisfaction.

H3=If students have more positive previous recommendations, then their satisfaction will increase.

Theoretical framework:



Methods section (methodology):

Elements of the research design:

Elements of the research design	Classification
Purpose of Study	Hypothesis Testing
Type of Investigation	Correlational with Causal Hypotheses
Extent of Researcher Interference	Minimal
Study Setting	Non-Contrived
Unit of Analysis	Individual Students
Time Horizon	Cross-Sectional

Justification:

Purpose of the study (hypothesis testing):

the study aims to test and validate various independent variables and their relationship with a dependent variable. It tests four sets of hypotheses regarding how students expectations, academic services, campus infrastructure and previous recommendations impact students satisfaction. By testing these variables and their relationships, the study seeks to confirm if the whole educational ecosystem such as academic challenges and service quality be valid for AASTMT students.

Type of investigation (Correlational with Causal Hypotheses):

the study is correlational because it examines how variables such as previous recommendations move in relation to students satisfaction. However it may tilt a little bit towards causal as it includes causal hypotheses by suggesting that improvements in specific independent variables (improved infrastructure) will directly lead to an increase in the dependent variable of student satisfaction.

Researcher interference (minimal):

to ensure that the collected data is as close reality as possible and the student experience is captured accurately, the research does not manipulate the environment of the study. Data is collected via questionnaires from students at AASTMT in their natural setting, observing the students perception, various opinions and point-of-view towards the whole educational ecosystem in the university.

Study setting (Non-Contrived):

the study takes place in a natural environment (AASTMT campus) where the educational process occurs daily and routinely. The study setting plays a crucial role because students satisfaction to the actual and physical experience of the campus infrastructure and real world interactions with academic services.

Unit of analysis (individuals):

the whole focus is on the individual because satisfaction is a personalized users belief and perceived quality. Each student unique experience with workload stress, personal recommendations from their family members and friends and individual employability goals serve as the primary source of data.

Time horizon (Cross-Sectional):

All data will be collected at a single time point, providing a snapshot of current relationships among variables. While the study mentions factors with long-term effects such as reputation and grade inflation, the primary data collection explores the current and present satisfaction levels and expectations of the students.

Population and sampling:

Target population: the targeted population consists of students from the Arab Academy for Science, Technology and Maritime Transport (AASTMT). Since the whole research was based on the level of students' satisfaction, this was the perfect match due to the availability of the data.

Sampling: the research required a total of responses ranging from 70 – 100 responses, that's why in our data collection we also targeted other branches and campuses of the university apart from the Smart Village campus. That was done not only to increase the number of responses but also to factor in diversity as we're studying the students' satisfaction in the whole university and not just one campus.

Data collection methods:

Research instrument:

the main method of data collection in this research were online questionnaires that were made using Google Forms, then it was distributed to friends across all the AASTMT branches.

Survey design:

the survey contains 6 sets of questions, divided into 2 sections. The first section containing questions that test all the independent variables in the research in addition to a set of questions that test the overall students' satisfaction level. The second question contains demographic questions that the responder should fill in.

Independent variables:

1. Students' expectations
2. Academic services
3. Campus infrastructure
4. Previous recommendations

Dependent variable:

1. Students' satisfaction

Demographics questions:

1. What's your age?
2. Gender
3. Marital status
4. Highest Completed Level Of Education
5. Academic Year
6. Enrollment Status
7. Average monthly Allowance (optional)

Measurement scale:

For each question there were 5 types of answers that the respondents could choose, it was divided into, strongly disagree, disagree, neutral, agree and strongly agree. Then when analyzing the responses, each one was coded into a number from 1 – 5.

- Strongly disagree = 1
- Disagree = 2
- Neutral = 3
- Strongly agree = 5
- Agree = 4

Interpretation/Discussion of the Result:

Frequencies:

			Statistics					
whats your age?			Gender	Marital Status	Highest Completed Level Of Education	Academic Year	Enrollment Status	Average monthly Allowance (optional)
N	Valid	76	76	76	72	72	72	52
	Missing	0	0	0	4	4	4	24

Frequency table:

whats your age?					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	under 20	21	27.6	27.6	27.6
	20 - less than 30	51	67.1	67.1	94.7
	above 30	4	5.3	5.3	100.0
	Total	76	100.0	100.0	

Gender					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	female	43	56.6	56.6	56.6
	male	30	39.5	39.5	96.1
	prefer no to say	3	3.9	3.9	100.0
	Total	76	100.0	100.0	

Marital Status					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	single	70	92.1	92.1	92.1
	married	6	7.9	7.9	100.0
	Total	76	100.0	100.0	

Highest Completed Level Of Education					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	bachelor's degree	30	39.5	41.7	41.7
	masters degree	5	6.6	6.9	48.6
	doctorate degree	1	1.3	1.4	50.0
	high school degree	36	47.4	50.0	100.0
	Total	72	94.7	100.0	
Missing	System	4	5.3		
Total		76	100.0		

Academic Year

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	first year	15	19.7	20.8	20.8
	second year	18	23.7	25.0	45.8
	third year	16	21.1	22.2	68.1
	fourth year	13	17.1	18.1	86.1
	postgraduate	10	13.2	13.9	100.0
	Total	72	94.7	100.0	
Missing	System	4	5.3		
Total		76	100.0		

Enrollment Status

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	local student	29	38.2	40.3	40.3
	international student	43	56.6	59.7	100.0
	Total	72	94.7	100.0	
Missing	System	4	5.3		
Total		76	100.0		

Average monthly Allowance (optional)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	less than 3000	9	11.8	17.3	17.3
	3000 - less than 5000	11	14.5	21.2	38.5
	5000 - less than 8000	13	17.1	25.0	63.5
	above 8000	19	25.0	36.5	100.0
	Total	52	68.4	100.0	
Missing	System	24	31.6		
Total		76	100.0		

Reliability scales:

Case Processing Summary

		N	%
Cases	Valid	76	100.0
	Excluded ^a	0	.0
	Total	76	100.0

a. Listwise deletion based on all variables in the procedure.

Scale: Previous Recommendations

Reliability Statistics

Cronbach's Alpha N of Items

.754	3
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Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item- Total Correlation	Cronbach's Alpha if Item Deleted
X1.1	7.64	2.685	.583	.671
X1.2	7.50	2.840	.610	.644
X1.3	7.41	2.645	.559	.700

Scale: Expectations**Reliability Statistics**

Cronbach's Alpha N of Items

.771	4
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Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item- Total Correlation	Cronbach's Alpha if Item Deleted
X2.1	11.55	4.357	.597	.707
X2.2	11.50	4.120	.618	.694
X2.3	11.80	4.161	.483	.770
X2.4	11.53	3.986	.612	.695

Scale: Perceived quality of service**Reliability Statistics**

Cronbach's Alpha N of Items

.751	3
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Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item- Total Correlation	Cronbach's Alpha if Item Deleted
X3.1	7.49	3.186	.610	.636
X3.2	7.57	2.756	.704	.516
X3.3	7.61	3.309	.445	.825

Scale: Infrastructures

Reliability Statistics

Cronbach's Alpha	N of Items
.896	5

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item- Total Correlation	Cronbach's Alpha if Item Deleted
X4.1	14.70	13.814	.703	.886
X4.2	14.45	14.811	.788	.864
X4.3	14.50	14.893	.757	.870
X4.4	14.51	14.120	.755	.870
X4.5	14.26	15.263	.740	.874

Scale: Student Satisfaction

Reliability Statistics

Cronbach's Alpha	N of Items
.855	4

Item-Total Statistics				
	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Y1.1	11.62	5.946	.732	.800
Y1.2	11.59	5.471	.734	.804
Y1.3	11.20	6.267	.706	.811
Y1.4	11.14	7.272	.649	.840

Correlations:

	total_recc	total_exp	total_qual	total_infr	total_satis
total_recc	1	.615***	.538***	.639***	.548***
total_exp	.615***	1	.712***	.573***	.606***
total_qual	.538***	.712***	1	.508***	.573***
total_infr	.639***	.573***	.508***	1	.743***
total_satis	.548***	.606***	.573***	.743***	1

Frequency analysis:

Age analysis:

The majority of the respondents with an approximate percentage of 67.1% are aged between 20 – 30, followed by 27.6% who are under 20 years of age. And only 5.3% are above 30. This indicates that the data sample primarily consist of university aged students.

Gender analysis:

56.6% of respondents identified as females, while 39.5% identified as male and only 3.9% preferred not to say. This may indicate that the factor of diversity was present while collecting the data from the targeted sample.

Marital status:

92.1% of the participants are single while 7.9% are married, this lines up perfectly with the age analysis showcasing that a wide variety of the students populations are actually young adults focused on their studies.

Educational background:

The highest being 50% of respondents having high school certificate as their highest completed level of education, then comes the bachelor degree with a 41.7%. the result indicates the mix between respondents those being as undergraduates and others of postgraduates.

Academic year:

Enrollment is relatively evenly spread across the first four years: first year (20.8%), second year (25.0%), third year (22.2%), and fourth year (18.1%). Postgraduates make up 13.9%. with most students being in their second year already, the authenticity of the data will be more reliable as by now they've must developed all their opinions and comments about the university.

Enrollment status:

International students dominates this by a percentage of 59.7 while local students achieving 40.3%, this spots the

light again on the factor of diversity as various opinions from different students coming from different backgrounds were collected.

Average Monthly Allowance:

With a percentage of 36.5%, students with an average monthly allowance of more than 8000, while 17.3% have an average monthly allowance of less than 3000. Its important to note that its for an allowance and not income which matches perfectly with what a student could spend on a month on average.

Reliability analysis:

Previous recommendations:

Cronbach's alpha = .754

The score indicates a good level of reliability with a score more than .6 . the items consistently measures how social influences, like various comments and opinions from the students inner circle, shaped the students view of the university. Because students view that pursuing a university degree as a high risk reward and a long-term investment, they relay on recommendations as precuations to reduce the risk of a wrong choice.

Students expectations:

Cronbach's alpha = .771

The score indicates a respectable level of consistency in the data scoring it above .6. the items reliabilty measures the benchmark that students set regarding the teaching skills and the overall staff quality in the university. The role of realibailty is important here cause it aims to see if the gap between expectations and reality eventually leads to satisfaction or disappointment.

Academic services:

Cronbach's alpha = .751

The score reached an acceptable reliability edge of .751 as it consistently measures the intangible factors of education such as the professors interest in the course and the teaching methods. Reliabilty scale is crucial here for the student as a customer (SAC) model, which suggests that in order for the students satisfaction to be high, the quality of service should be high as well.

Campus infrastructure;

Cronbach's alpha = .896

This sacle has registered the highest score on all the study with a score of .896, approaching an excellent rating. It very consistently measures the tangible factors such as classrooms facilities and modern study places. This high consistency suggests that students perceive their physical environment as a unified academic ecosystem or in other words as the main source of their satisfaction. The study justifies this high focus because poor infrastructures act as a limiting factor that holds back the students potential.

Students satisfaction:

Cronbach's alpha = .855

This score of .855 indicates high reliability. It confirms that the questions regarding the overall experience and the decision to choose AASTMT are measuring a single, stable belief shared among most students in all campuses. Since satisfaction is the "most important KPI" for the university, having a highly reliable scale ensures the study's conclusions about what drives student happiness are statistically correct.

Correlations analysis:

Relationship with students satisfaction (the dependent variable):

The main goal was to see what drives students satisfaction. All variable show a moderate to strong positive relationship with satisfaction.

Students expectations – students satisfaction (0.606) = strong positive relationship.

Among all the other variables, this was the highest correlation towards students satisfaction. It indicates that when universities align with or exceeds the initial benchmark set by students, in return the students satisfaction level will increase significantly.

Academic quality – students satisfaction (0.573) = moderate to strong positive relationship

There's a clear link between the intangible services and how happy the student is with their choices.

Previous recommendations – students satisfaction (0.548) = moderate positive relationship

Students who enrolled into the university based on positive word-of-mouth from their inner circles tend to report higher satisfaction levels, likely because their expectations weren't just met but also pre-approved by trusted sources.

Campus infrastructure – students satisfaction (0.522) = moderate positive relationship

While still being significant, the physical environment has a lower direct correlation with satisfaction than other academic factors, indicating that while facilities matter, they're secondary to the quality of teaching and expectations.

Limitations of the study:

Geographic sample size: while the study tries to collect many responses ranging from 70 – 100 response by including other branches on the AASTMT and to also increase the diversity in the participants responses, however the sample size remains still relatively small compared to a sample size that must have covered a whole university and its branches across the country.

Cross-sectional time horizon: the data was collected only once which provided a snapshot of the level of students satisfaction or their expectations that may evolve over time due to various reasons such as the evolve of the courses or full degree or even the weather.

Selection bias: the questionnaires were distributed to friends across AASTMT branches, this method of distribution can lead a selection bias were the respondents might share similar viewpoints and backgrounds that do not fully represent the diverse demographics of the whole students population.

Conclusion:

Comparison with previous research:

The research is aimed specifically at identifying internal and external factors that influence the overall students satisfaction at the arab academy for science, technology and maritime transport. While the journal study takes a broader and more of a macro approach to the situation as the research doesn't stop when the students satisfaction is fulfilled but it continues to using that students satisfaction in order to foster long-term sustainability for higher education institutions. The AASTMT study examines various factors such as the students expectations, academic services, campus infrastructure and previous recommendations to test and reach the final goal of students satisfaction. The journal study explore a broader and much wider contrast by mentioning students satisfaction as a variable alongside with other variable that are used to foster for long-term sustainability. In short, the AASTMT mentions and try to solve minor or in other terms micro problem areas associated with students satisfaction where as the journal study tries to sole broader or macro problem areas while using students satisfaction as a variable.

Recommendation for further research:

For the future research following the AASTMT study, its recommended to following a longitudinal time horizon for data collection instead of using a cross-sectional. As its crucial to track not only the students expectations and students satisfaction but its evolution overtime during all the academic years the students been through from freshman to a graduate. Since the current research mentions the role of experience in assisting students I matching their judgments to their expectations, a longitudinal approach will reveal how students shift their perception of reality as they become more integrated into the university environment. In addition, regarding the sample size, it will be much better to increase the size of the targeted sample in order to increase the diversity rate in the responses which will in return will provide more reliable data and a better understanding of the whole relationship between the independent variables and the students satisfaction. That will be a great opportunity as the time-horizon of the data collection process will be shifted from a cross-sectional to a longitudinal one. Lastly, the data distribution methods should be more precise, when targeting the students population samples on all of AASTMT university campuses, choose a sample in each campus that contain diversity just to make sure that there are no any sort of selective

biases later on in the responses. All points mentioned above will be better synced with the longitudinal time horizon .

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Appendix:

Frequencies

		Statistics						
		whats your age?	Gender	Marital Status	Highest Completed Level Of Education	Academic Year	Enrollment Status	Average monthly Allowance (optional)
N	Valid	76	76	76	72	72	72	52
	Missing	0	0	0	4	4	4	24

Frequency Table

		whats your age?			
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	under 20	21	27.6	27.6	27.6
	20 - less than 30	51	67.1	67.1	94.7
	above 30	4	5.3	5.3	100.0
	Total	76	100.0	100.0	

Gender

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	female	43	56.6	56.6	56.6
	male	30	39.5	39.5	96.1
	prefer no to say	3	3.9	3.9	100.0
	Total	76	100.0	100.0	

Marital Status

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	single	70	92.1	92.1	92.1
	married	6	7.9	7.9	100.0
	Total	76	100.0	100.0	

Highest Completed Level Of Education

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	bachelor's degree	30	39.5	41.7	41.7
	masters degree	5	6.6	6.9	48.6
	doctorate degree	1	1.3	1.4	50.0
	high school degree	36	47.4	50.0	100.0
	Total	72	94.7	100.0	
Missing	System	4	5.3		
Total		76	100.0		

Academic Year

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	first year	15	19.7	20.8	20.8
	second year	18	23.7	25.0	45.8
	third year	16	21.1	22.2	68.1
	fourth year	13	17.1	18.1	86.1
	postgraduate	10	13.2	13.9	100.0
	Total	72	94.7	100.0	
Missing	System	4	5.3		
Total		76	100.0		

Enrollment Status

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	local student	29	38.2	40.3	40.3
	international student	43	56.6	59.7	100.0
	Total	72	94.7	100.0	
Missing	System	4	5.3		
Total		76	100.0		

Average monthly Allowance (optional)

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	less than 3000	9	11.8	17.3	17.3
	3000 - less than 5000	11	14.5	21.2	38.5
	5000 - less than 8000	13	17.1	25.0	63.5
	above 8000	19	25.0	36.5	100.0
	Total	52	68.4	100.0	
Missing	System	24	31.6		
Total		76	100.0		

Reliability

Scale: Previous Recemmendations

Case Processing Summary

		N	%
Cases	Valid	76	100.0
	Excluded ^a	0	.0
	Total	76	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.754	3

Item Statistics

	Mean	Std. Deviation	N
X1.1	3.63	.964	76
X1.2	3.78	.888	76
X1.3	3.87	.998	76

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
X1.1	7.64	2.685	.583	.671
X1.2	7.50	2.840	.610	.644
X1.3	7.41	2.645	.559	.700

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
11.28	5.456	2.336	3

Reliability

Scale: Expectations

Case Processing Summary

		N	%
Cases	Valid	76	100.0
	Excluded ^a	0	.0
	Total	76	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.771	4

Item Statistics

	Mean	Std. Deviation	N
X2.1	3.91	.769	76
X2.2	3.96	.824	76
X2.3	3.66	.932	76
X2.4	3.93	.869	76

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
X2.1	11.55	4.357	.597	.707
X2.2	11.50	4.120	.618	.694
X2.3	11.80	4.161	.483	.770
X2.4	11.53	3.986	.612	.695

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
15.46	6.865	2.620	4

Reliability

Scale: Perceived quality of service

Case Processing Summary

		N	%
Cases	Valid	76	100.0
	Excluded ^a	0	.0
	Total	76	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.751	3

Item Statistics

	Mean	Std. Deviation	N
X3.1	3.84	.953	76
X3.2	3.76	1.018	76
X3.3	3.72	1.066	76

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
X3.1	7.49	3.186	.610	.636
X3.2	7.57	2.756	.704	.516
X3.3	7.61	3.309	.445	.825

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
11.33	6.170	2.484	3

Scale: Infrastructures

Case Processing Summary

		N	%
Cases	Valid	76	100.0
	Excluded ^a	0	.0
	Total	76	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.896	5

Item Statistics

	Mean	Std. Deviation	N
X4.1	3.41	1.288	76
X4.2	3.66	1.040	76
X4.3	3.61	1.059	76
X4.4	3.59	1.180	76
X4.5	3.84	1.020	76

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
X4.1	14.70	13.814	.703	.886
X4.2	14.45	14.811	.788	.864
X4.3	14.50	14.893	.757	.870
X4.4	14.51	14.120	.755	.870
X4.5	14.26	15.263	.740	.874

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
18.11	22.202	4.712	5

Reliability

Scale: Student satisfaction

Case Processing Summary

		N	%
Cases	Valid	76	100.0
	Excluded ^a	0	.0
	Total	76	100.0

a. Listwise deletion based on all variables in the procedure.

Reliability Statistics

Cronbach's Alpha	N of Items
.855	4

Item Statistics

	Mean	Std. Deviation	N
Y1.1	3.57	1.011	76
Y1.2	3.59	1.122	76
Y1.3	3.99	.959	76
Y1.4	4.04	.774	76

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
Y1.1	11.62	5.946	.732	.800
Y1.2	11.59	5.471	.734	.804
Y1.3	11.20	6.267	.706	.811
Y1.4	11.14	7.272	.649	.840

Scale Statistics

Mean	Variance	Std. Deviation	N of Items
15.18	10.579	3.253	4

Correlations

		Correlations				
		total_recc	total_exp	total_qual	total_infr	total_satis
total_recc	Pearson Correlation	1	.615 ^{***}	.538 ^{***}	.639 ^{***}	.548 ^{***}
	Sig. (2-tailed)		<.001	<.001	<.001	<.001
	N	76	76	76	76	76
total_exp	Pearson Correlation	.615 ^{***}	1	.712 ^{***}	.573 ^{***}	.606 ^{***}
	Sig. (2-tailed)	<.001		<.001	<.001	<.001
	N	76	76	76	76	76
total_qual	Pearson Correlation	.538 ^{***}	.712 ^{***}	1	.508 ^{***}	.573 ^{***}
	Sig. (2-tailed)	<.001	<.001		<.001	<.001
	N	76	76	76	76	76
total_infr	Pearson Correlation	.639 ^{***}	.573 ^{***}	.508 ^{***}	1	.743 ^{***}
	Sig. (2-tailed)	<.001	<.001	<.001		<.001
	N	76	76	76	76	76
total_satis	Pearson Correlation	.548 ^{***}	.606 ^{***}	.573 ^{***}	.743 ^{***}	1
	Sig. (2-tailed)	<.001	<.001	<.001	<.001	
	N	76	76	76	76	76

***. Correlation at 0.001 (2-tailed)